2 **A1** Procedure OK (i.e. measure pressure and electrical power; change pressure and measure new power and repeat). **[1]**
This mark can be scored even if the method is unworkable.

A2 Diagram of workable arrangement **[2]**
e.g. wire connected to joulemeter and power supply. A stopwatch must be used if this method is employed. Allow alternative circuit using ammeter and voltmeter.
The wire must be shown inside a closed container with some means of changing and measuring the pressure.
One mark for the electrical arrangement; one mark for the mechanical arrangement.

A3 Measurement of pressure **[1]**
(e.g. Bourdon gauge/pressure gauge/manometer)

B1 Change setting on the power supply to keep the wire at the same temperature when the pressure is changed. **[1]**

B2 Power supplied to wire = V x I (or reading on joulemeter ÷ time) **[1]**

B3 Use of thermocouple thermometer to monitor temperature whilst pressure is changed or adjust V or I to keep R (and hence T) constant **[1]**

C Any one safety precaution **[1]**
e.g. safety screens/goggles/wire mesh surrounding vacuum chamber

D1/2 Any further good design features **[2]**
Some of these might be:
Thermocouple thermometer shown attached to the wire to monitor temperature.
Light spot galvanometer connected to thermocouple.
Use of a needle valve to control pressure.
Vacuum grease the connecting wires to the heater wire.
Allow time between readings for experiment to stabilise.
Do not allow the wire to become too hot or the thermocouple may melt.

10 marks in total

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